

Yulong Liu

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RESEARCH INTERESTS

AI methods & discovery: Differentiable programming, hybrid neural PDE solvers, uncertainty-aware scientific AI, and AI for Science and scientific discovery.

Engineering: Homogenization and multiscale modeling, coupled multiphysics simulation, and subsurface energy systems, including geothermal reservoirs and hydraulic fracturing.

EDUCATION

Ph.D. in Earth Science – Cornell University 2024 – Present

Minors: Computer Science and Scientific Computing

B.S. in Mining Engineering – Northeastern University (CN) 2024

RESEARCH EXPERIENCE

Ph.D. Candidate – Dept. of Earth and Atmospheric Sciences, Cornell University Aug. 2024 – Present

Advisor: Chloé Arson

Ithaca, NY

Project: Scientific AI and scientific discovery for multiscale, multiphysics geomechanics and subsurface energy systems.

- Develop coupled thermo-hydro-mechanical models in MOOSE for subsurface energy systems, including geothermal reservoirs and hydraulic fracturing.
- Study differentiable programming and hybrid neural PDE solvers, combining physics-informed networks, implicit neural representations, and neural operators with numerical simulation.
- Investigate uncertainty-aware surrogate modeling for geothermal systems, with an emphasis on uncertainty propagation and the reliability of multiphysics predictions.
- Lead a cross-institution project on PDE-grounded verification of the physical correctness of LLM-generated multiphysics simulation code.
- Study homogenization and multiscale transport in heterogeneous porous media; use AI for scientific discovery to learn interpretable effective permeability laws.

Undergraduate Researcher – Northeastern University (CN)

Sep. 2020 – Jun. 2024

B.S. in Mining Engineering

Shenyang, China

Thesis: A Unified Model for Microseismic Signal Classification and Arrival-Time Picking Based on Deep Learning

- Developed a unified deep-learning model for microseismic signal classification and arrival-time picking, integrating tasks typically handled in separate workflows.
- Conducted experimental and computational research on hydraulic fracturing, triaxial rock testing, and tunnel boring machines; received the Outstanding Thesis Award and National Scholarship.

GRANTS, PROPOSALS, AND COMPUTATIONAL AWARDS

Learned Uncertainty-Propagation and World Models for Enhanced Geothermal Reservoirs 2026

National Science Foundation (NSF), ACCESS program. Student lead: Y. Liu; Advisor: C. Arson. Total allocation and Arson group's share: 1.5M CPU hours.

AI-aided Computational Geomechanics

2026

Empire AI Alpha+Beta. PIs: Y.C. Han, Y. Liu, A. Tristani, and C. Arson. Total allocation and Arson group's share: 5.5K GPU hours.

NASA FINESST Proposal – Title Withheld During Anonymous Review

2026

NASA ROSES-25 F.5 FINESST. Under review. Future Investigator: Y. Liu; PI and faculty mentor: C. Arson.

PUBLICATIONS

Selected Papers

- **Yulong Liu**, Jonah Botvinick-Greenhouse, Yunan Yang, Chloé Arson. *Operator Learning Surrogate Modeling of Hydraulically Fractured Geothermal Injection–Production Systems: A Cornell Case Study*. ARMA US Rock Mechanics/Geomechanics Symposium, Tucson, AZ, 2026.
- **Liu, Y.**, Arson, C. *A Physics-Informed Neural Network for Pressurized Cavities of Arbitrary Shape in Heterogeneous Rock*. *Rock Mech Rock Eng* (2026). doi:10.1007/s00603-026-05882-5.
- M. Belachew, **Yulong Liu**, J. D. Frost, Chloé Arson. *Numerical Assessment of Plasticity Development and Energy Expenditure of Ant-Like Microtunnelling*. *Tunnelling and Underground Space Technology*, 172, 107501, 2026.
- **Yulong Liu**[†], Chloé Arson. *Physics-Informed Neural Network Surrogate Modeling of Pressurized Cavity in Homogeneous and Bilayered Media*. ARMA US Rock Mechanics/Geomechanics Symposium, D022S018R006, 2025.
- Zhenghan Song, **Yulong Liu**^{*†}, et al. *Your Simulation Runs but Solves the Wrong Physics: PDE-Grounded Intent Verification for LLM-Generated Multiphysics Simulation Code*. [arXiv:2605.09360](https://arxiv.org/abs/2605.09360). Under review at NeurIPS 2026.
- Zhenghan Song, Yunyi Li, **Yulong Liu**[†]. *Prefix-Safe Bayesian Belief Tracking for LLM Reasoning Reliability: Separating Calibration from Ranking*. [arXiv:2605.27712](https://arxiv.org/abs/2605.27712). Under review at ACL ARR 2026.

Thesis

- **Yulong Liu**. *A Unified Model for Microseismic Signal Classification and Arrival-Time Picking Based on Deep Learning*. Undergraduate thesis, Northeastern University, 2024.

* Co-first author † Corresponding author

CONFERENCES AND PRESENTATIONS

Oral presentation, 60th US Rock Mechanics/Geomechanics Symposium (ARMA) <i>American Rock Mechanics Association, Tucson, AZ, USA</i>	Jun. 2026
Invited talk, 20th Phase Field Methods Workshop <i>Center for Hierarchical Materials Design, Northwestern University, Evanston, IL, USA</i>	May 2026
Poster presentation, Cornell CEE Graduate Research Symposium <i>Cornell University, Ithaca, NY, USA</i>	Apr. 2026
Invited seminar: “Physics-Informed AI Optimization of Heterogeneous Rock Systems” <i>Southwest Petroleum University seminar series, Chengdu, China</i>	Dec. 2025
Poster presentation, 59th US Rock Mechanics/Geomechanics Symposium (ARMA) <i>American Rock Mechanics Association, Santa Fe, NM, USA</i>	Jun. 2025

FELLOWSHIPS AND AWARDS

Estwing Hammer Award – Most Outstanding EAS Graduate Student of 2024, Cornell University	2025
Cornell University Travel Grant , Cornell University	2025, 2026
Outstanding Thesis , Northeastern University	2024
National Scholarship , Ministry of Education, China	2024
First Class Scholarship , Northeastern University	2023
Autumn Scholarship , Northeastern University	2022
Golden Seed Scholarship , Northeastern University	2022

ACADEMIC SERVICE

- Reviewer, *Rock Mechanics and Rock Engineering*
- Reviewer, *Underground Space*
- Reviewer, *International Conference on Artificial Neural Networks*
- Reviewer, *Geothermal Rising Conference*

SKILLS

Programming & Scientific Computing: Python, C++, R, MOOSE finite element framework

Modeling & Simulation: finite element modeling, coupled multiphysics simulation, poromechanics, computational geomechanics, CAD (e.g., AutoCAD, SolidWorks)

Subsurface & Reservoir: geothermal reservoirs, hydraulic fracturing, injection–production modeling, homogenization and multiscale transport in porous media

Scientific Machine Learning: differentiable programming, hybrid neural PDE solvers, uncertainty-aware scientific AI, AI for Science and scientific discovery; physics-informed networks, neural operators, and implicit neural representations

LLMs for Simulation: using large language models to accelerate simulation-software workflows and to verify the physical correctness of generated simulation code

Experimental Methods: rock mechanics testing (e.g., triaxial compression), laboratory instrumentation

Currently Learning: phase-field modeling (fracture/multiphysics), multiphase flow in porous media

Scientific Writing: L^AT_EX / Overleaf

PROFESSIONAL MEMBERSHIPS

- Member, American Rock Mechanics Association (ARMA), 2025–Present
- Member, International Society for Rock Mechanics and Rock Engineering (ISRM), 2025–Present
- Member, Chinese Society for Rock Mechanics and Engineering (CSRME), 2024–Present
- Member, Geothermal Rising, 2025–Present